CS174A Team Project

General Info

- Team sizes: 3
- Projects assignments 1-4 should provide you the background needed for your project
- Project must include basic topics of syllabus at least through 7th week of quarter
- Each team is expected to implement interactive graphics and/or mouse picking + min 2 advanced features, as described below and discussed with your TA
- Most stories are inspired by existing games, extend to 3D
- You can use 3js (modeling objects, bump maps, particles, etc.) and OBJ loaders, but no other external libraries or frameworks are allowed; see examples folder for ideas, e.g., for placing 2D/3D texts in your scene
- Project discussions will occur during Friday TA sessions

Advanced Features

- Ray tracing
- Shadowing
- Collision detection
- Physics-based simulation
- Bump mapping/displacement mapping
- Artificial intelligence
- Scientific data or simulation visualization

Some references for project ideas:

- Use LLMs like ChatGPT, GPT-4, Gemini, Copilot, etc. to get creative ideas for your project
  Example prompt: Act as a student of college-level 3D computer graphics course. Give me ideas for a team project that requires lot of coding/programming; project should have user interactivity using mouse or keyboard; project should have some advanced features like <include from above list based on your interest>; modify prompt based on your story.
- Aug 2024 SigGraph Conference
- Collision Detection: AABB and SAT methods
- Autonomous Vehicles: 14-16 mins for 3D bounding boxes collision detection (DeepRoute's AI algo, using Nvidia's GTC)
- Flocking Birds: no collision, following path/pattern set by modeler
- Disney+ → NatGeo → Welcome to Earth → Episode 3, Mind of the Swarm: not too close, not too far, follow your neighbor
- MedVR: VR env on top of WebGL for training medical students, physics-based interactions
- Avatar Motion Capture
- Coca Cola Ad

Examples

From previous quarters: http://web.cs.ucla.edu/~dt/courses/CS174A/animations/

Initial Project Proposal

Content

- Approximately 1-2 pages
- Include project name, name of team members, emails, student IDs
- What is the theme (or story) of your animation? Include pictures, animations, details
- What topics learnt in the course are used and how?
- What interactivity will you use?
- What advanced features will you be implementing?
- Provide semi-technical commentary about how you'll be implementing these features
- Easy to read; no formal guidelines, just tell us what your idea is in the best possible manner

Suggestions for Writing a Good Proposal

- **Before you finalize your proposal:**
  You can have a quick talk with your TA to ensure you are in the right direction, and what you are thinking makes sense as a project for this course (optional but highly recommended).
- **Goal #1 of proposal:**
  The first goal of the project proposal is to clearly and unambiguously describe what you plan on doing. For this reason, you should stick to clear and concise language. Pictures or drawings can help understanding. It is fine if you do not know exactly how much of it you will have time to do. Explicitly distinguish between what features you are sure you will implement and what features you might implement if you have enough time.
- **Goal #2 of proposal:**
  Second goal is for us (and for you) to assess the feasibility of the project. For this reason, if you have advanced features, make sure you have some description of how you want to implement it, enough for us to see you know what needs to be done. For example, if you have collision detection, briefly explain how you will approximate object boundaries, detect the collision, and what happens if two objects collide. If you want to do a physics simulated animation, describe what physical phenomena you will be modeling. These descriptions can be short, semi-technical.

IMPORTANT: You can use GenAI tools (MS Copilot, Google Gemini, OpenAI ChatGPT) to evaluate and improve your scores; but you **CANNOT** use these tools to write the entire report or to do peer evaluations or to generate all your code…that would be considered **PLAGIARISM**. TAs will also use one of these tools to evaluate your proposals but will modify/add/override with their own scores and comments.

Initial Proposal/Demo Rubric

- Creativity
- Features
- Proposal
- Presentation/Demo
- Progress Made

Final Project Report & Code

Report

- Approximately 5-6 pages
- Should be a refined version of initial proposal
- Should include feedback provided by TAs and peers
- Should be in "final" form, TAs and peer evaluations will expect what you have promised in this final version
- Bring your proposals approximately in par with what most others are doing; it should not be too less or much more than other projects
- Take care of all gaps and to-do's that you had noted in initial version
- Usage Instructions: Explain your project that a 3rd person can understand and execute, like goal, features, how to operate, what do arrow keys do (or WASD keys), what do mouse buttons do (mouse picking), how to score points in your game, how to get to the next level, sounds effect if any, etc.
- Lessons learned, challenges faced; how did you overcome them
- What was your team's contribution on top of using LLMs for generating your initial idea or code
- How you liked peer eval process for evaluating, feedback, and TM evaluations
- What more could you have done had you had more time
- What other suggestions you can provide for improving the process

Code

- Ensure code is runnable/executable without compilation errors
- Code should be consistent with your report and demo
- Code will be used to assess technical depth and features implemented
- Code should not have been plagiarized

Final Report/Demo Rubric

- Features Implemented/Completeness
- Quality of Report
- Presentation/Demo
- Usage Instructions
- Lessons Learned & Next Steps

Peer Evaluation & Rubrics

General Instructions

- TAs will grade initial proposal and provide feedback to project team
- TAs will assign 2 projects for each student to evaluate
- You will act as an instructor and be responsible for evaluating the quality of your peers' projects
- You are expected to evaluate initial project proposals and demo, look for consistency between the project's proposal and what they delivered, team work, level of effort, etc.
- Your evaluation should be motivational as well as critical; be honest, specific, and kind
- Your review and feedback is considered complete only after you provide a rating for each criterion in the rubric
- The project team will provide feedback on your evaluation based on "helpfulness" (detailed, thoughtful, specific, quality); goal is to improve their work

Rating Scale

All ratings will be on a scale from 1-10 (1-2: needs work, 3-4: fair, 5-6: good, 7-8: great, 9-10: exemplary)
In addition, you will add a comment indicating why you gave a particular rating

Team Member Evaluation Rubric

- Team Work
- Communication
- Content Knowledge & Skills
- Work Ethic
- Overall Contribution

Initial & Final Demos

General Instructions

- Each team will have 5-6 mins to present + 2 mins for Q&A + switching time (e.g., unable to present or share screen)
- All team members should individually speak about their respective contributions during demos
- Presenters should turn on their videos while presenting
- When one person presents, others in the room should place themselves on mute to avoid echoes
- Rehearse your presentation and demo so that all team members are on the same page
- Use “optimize for video clip” and “share sound” to make your animation plays smoothly
- Peer evaluations should read proposal beforehand and listen during demo; provide your evaluation in P, it’s anonymous and not graded
- Ask questions, especially if you’d like to know how a team implemented something that you might want to include in your project
- Compare your project with what other teams have implemented, and adjust your proposal accordingly, including TA/peer’s feedback
- See the latest demo-related announcement(s) that may have been posted in Canvas, like sign-up sheets, zoom links, presentation orders, peer evaluation instructions, etc.
- In the spirit of a 3D graphics course, make your scenes visually appealing
- Demos for all sessions will be recorded and posted on Canvas; if you’re evaluating a project that’s being presented in another session, you will be able to watch the video offline and provide your feedback

Presentation Template

- 2-3 presentation slides
- Name of project and team members
- Storyline of project, goal, etc.
- Features implemented, including advanced features + interactivity
- What is implemented so far (for initial demo)
- What more will the peers see during final demo (for initial demo)
- What challenges did you face, and how did you resolve them
- Don’t present code, but technically explain your features
- Q&A

Initial Demo Expectations

- All characters should’ve been modeled, loaded, and transformed to form the environment, using Blender/Maya, ObjLoader, Shaders
- Keyboard/mouse navigations should be functional
- Prototype of your advanced features

Final Demo Expectations

- All remaining features that’re promised in your proposal
- Features that were covered in class or discussions after initial demo, like shadows, textures, collisions, physics simulations, and other advanced features
- Music or sound integrations, other constraints

Grading & Timelines

Grading (175 points = 35% of total):

- Initial proposal + demo: 5%
- Peer evaluations (2 per student) + evaluation feedback (4-6 per project): 10%
- Final code + report + demo: 15%
- Team member evaluation: 5%

Project Timelines using Peerceptiv (P)

Date	%	Students	TAs
2/09		Form groups in P, with informative team names	
2/18	3	**Submission**: Upload initial proposal, one per group in P	Assign each student 2 projects to peer eval in P
2/20	2	**Demo**: Initial demo live in discussion class	
2/25	2+4	**Evaluation**: Peer evaluations in P	TA evaluations in P (initial proposals/demos)
2/27	2+2	**Feedback**: Provide feedback on your peer evals	
3/08			Create final demo assignment in P
3/13		**Demo**: Final demo live in discussion class	
3/14		**Submission**: Zip and upload code in BL	
3/20		**Submission**: Upload final report in P	
3/23	2+3	**Feedback**: Provide feedback on your team members	
3/24	15		TA evaluates in P (final code, demos, and reports)
3/27			Export all grades from P to Canvas

Additional Notes

- Only team leader will submit in P: initial proposal, final report, code
- Only 1 doc can be uploaded per assignment (zip files if there're multiple files)
- Upload only PDF versions of your proposals
- Peer evaluation process will be anonymous, but TAs will add "TA" in the comments to indicate which evaluation came from TA
- You may be assigned projects from other sections to evaluate, so view the demo recordings once they're posted on Canvas, at least the ones that you're evaluating

Peerceptiv Help Materials

Introduce Peerceptiv to Students
Student Videos
Student Help Articles
Understanding Peerceptiv Results